ERTH555 Problem Set 2

Zac Cross

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Problem 1

$$\sigma_{xx} = -40\text{MPa}, \sigma_{xy} = -60\text{MPa}, \sigma_{yy} = -100\text{MPa} \tag{1}$$

a. Find the components of the outward unit normal vector, $n_x = \cos\theta$ and $n_y = \sin\theta$, in the prescribed coordinate system and use them to compute tx(n) and ty(n), the x- and y-components of the traction vector acting on the fault surface. From the picture, the angle θ between the normal and the x-axis is about 120 degrees.

Given $\theta = 120°$,

$$\hat{n} = \cos(120)\hat{x} + \sin(120)\hat{y} \tag{2}$$

$$= -\frac{1}{2}\hat{x} + \frac{\sqrt{3}}{2}\hat{y}, \tag{3}$$

we can calculate the compents of the traction $\boldsymbol{t}$ as follows:

$$t_i(n) = \sigma_{ij}n_j, \text{ so} \tag{4}$$

$$t_x = \sigma_{xx}n_x + \sigma_{xy}n_y \tag{5}$$

$$= -40(-\frac{1}{2}) + -60(\frac{\sqrt{3}}{2}) \approx -31.96, \tag{6}$$

$$t_y = \sigma_{yx}n_x + \sigma_{yy}n_y \tag{7}$$

$$= -60(-\frac{1}{2}) + -100(\frac{\sqrt{3}}{2}) \approx -51.60, \tag{8}$$

$$\tag{9}$$

so the traction vector $\boldsymbol{t(n)} = -31.96\hat{x} - 51.60\hat{y}$ with units of MPa.

b. Calculate the magnitude and direction of the traction (measured in degrees counterclockwise from the x-axis), t(n), acting on the fault surface with outward normal n.

The magnitude of **t(n)** is given by the sum of the square of the components to be:

$$||\boldsymbol{t(n)}|| = \sqrt{t_x^2 + t_y^2} \tag{10}$$

$$= \sqrt{(-40(-\frac{1}{2}) + -60(\frac{\sqrt{3}}{2}))^2 + (-60(-\frac{1}{2}) + -100(\frac{\sqrt{3}}{2}))^2} \tag{11}$$

$$= 65.00 \text{ MPa} \tag{12}$$

the direction θ' of the traction from the x axis is given by:

$$\theta' = \tan^{-1}(\frac{t_y}{t_x}) + 180^\circ \tag{13}$$

$$= \tan^{-1}(\frac{-60(-\frac{1}{2}) + -100(\frac{\sqrt{3}}{2})}{-40(-\frac{1}{2}) + -60(\frac{\sqrt{3}}{2})}) + 180^\circ \tag{14}$$

$$= 60.55 + 180 = 240.55^\circ \tag{15}$$

where the tan term comes from the trig of vector addition with a 180 ° offset because both components are in the 3rd quadrant of the x-y plane.

c. Draw a carefully labeled sketch map of the fault surface with normal vector n, the traction vector t(n), and the Cartesian coordinate system. Use your sketch to indicate if this traction pushes or pulls against that surface. Remember it represents the mechanical action of the rock in the upper left part of the photo on the surface bounding the rock of the lower part of the photo

See the figure below for the sketch. We can see that the upper left rock is pushing "down" on the lower right rock in the -x and -y directions.

Figure 1: Sketch of fault in Cartesian Coordinates

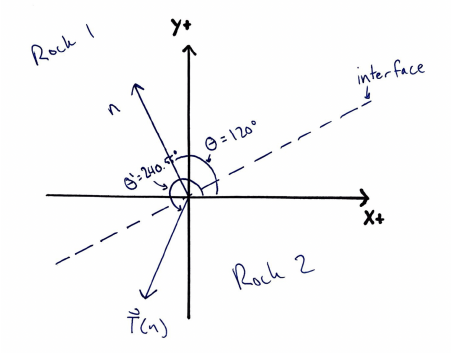
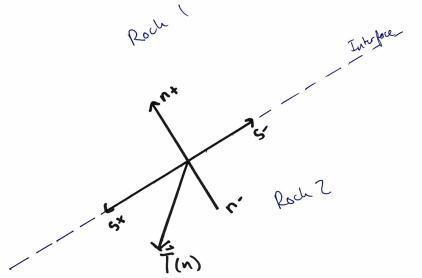


Figure 2: Sketch of fault in with normal and shear axis



d. Now redraw the sketch and add the coordinate system representing the axes that are normal n and tangential s to the fault surface. Calculate the normal and shear traction components on the fault surface. Is the sign of the shear traction component consistent with the sense of shearing indicated by the offset dike in Figure 1? Explain your reasoning.

The new sketch can be see in figure 2. We can calculate the normal component of the traction by projecting our traction result onto the normal vector:

$$
\begin{aligned}
t_n &= (t(n) \cdot \hat{n})\hat{n} && (16)\\
&= (t_x n_x + t_y n_y)\hat{n} && (17)\\
&= -33.04\hat{n} && (18)\\
& && (19)
\end{aligned}
$$

The shear component can be found by projecting the traction onto the shear

unit vector. The shear direction is at 90 ° offset from the normal, since this is a 2D problem, so we can solve for the shear component as:

$$
\begin{aligned}
t_s &= (t(n) \cdot \hat{s})\hat{s} && (20)\\
&= (t_x s_x + t_y s_y)\hat{s} && (21)\\
&= (t_x \cos(210) + t_y \sin(210))\hat{s} && (22)\\
&= 55.98\hat{s} && (23)
\end{aligned}
$$

where $\hat{s} = \cos(210)\hat{x} + \sin(210)\hat{y}$.

I think these directions make sense with the picture of the Dike in figure one. It seems to form in the +s and -n directions, matching the signs we got above.

Problem 2

a. For dikes to form at the edge of a hypothetical cylindrical magma chamber, the tangential stress component $\sigma_{\theta\theta}$ needs to reach and exceed the tensile strength of the rock. Where (at whatθ) will tensile tangential stress be maximized along the edge of the chamber? Find an equation for the minimum magma pressure P, in terms of SH and Sh, required to initiate tensile fracture at the edge of the chamber (this occurs when $\sigma_{\theta\theta} = T$ where T is the tensile strength of the rock)

Looking at the expressions for $\sigma_{\theta\theta}$, we can see that the third term is the only one that depends on θ. Since this term is strictly 0 or Positive, it is maximized when $\cos(2\theta) = 1$, and r = R. So, $\theta = 0, \pi$.

We can now find an expression for the minimum magma pressure P in terms of S_H, S_h at these points by setting $\cos(2\theta)$ to 1 and setting the whole term to T as follows:

$$T = \sigma_{\theta\theta} \tag{24}$$

$$= -\frac{1}{2}(S_H + S_h)(1 + (R/r)^2) + P_{min}(R/r)^2 + \frac{1}{2}(S_H - S_h)(1 + 3(R/r)^4)\cos(2\theta) \tag{25}$$

$$= -\frac{1}{2}(S_H + S_h)(2) + P_{min} + \frac{1}{2}(S_H - S_h)(4) \tag{26}$$

$$= (-S_H + -S_h) + P_{min} + (2S_H - 2S_h) \tag{27}$$

$$T = (S_H - 3S_h) + P_{min} \tag{28}$$

Moving the P over, we get the final expression:

$$P_{min} = T + 3S_h - S_H. \tag{29}$$

b. Imagine the magma chamber is at 4 km depth, where magnitudes of the regional horizontal stresses are SH = 125 MPa and Sh = 100 MPa. Tensile strength of rock is only 1 – 10 MPa. For these conditions, what value of magma pressure P is required to initiate a dike in an otherwise unfractured rock? Would you expect magma pressure ever to achieve this value based on the geological literature? Explain your reasoning.

Given the values above, our P_{min} would have to be between

$300 - 125 + 1 = 176$ MPa at the low end of Tensile strength and $300 - 125 + 10 = 185$MPa at the high end of tensile strength.

Doing a quick google and skim of articles and research literature in the area, it looks like a magma pressure on the order or 100-200 MPa seems reasonable. This paper in AGU Caracausi, et al. reported pressures up to 400 MPa by estimating magma pressure using degassing measurements.

c. Set the magnitude of magma pressure to 175 MPa, and vary the difference between the two remote principal stresses (sH and sh). Describe qualitatively the effect on the stress trajectory pattern. What can one learn about the remote state of stress from the orientations of dikes at significant distances from a magma chamber? What general conclusions would you draw if the map pattern of dikes at a certain volcanic vent are purely radial (like spokes)?

Looking at the figures below for various values of S_H and S_h, we can see that as the difference between the two remote principal stresses gets larger, the S_H term begins to dominate the dike trajectory, forcing a horizontal flow. As we make the remote principal stress terms larger than the pressure, we also see that they dominate the flow even near the magma chamber. However when S_H and S_h are the same order of magnitude as the pressure and the difference is small, we see a radial behavior. We see similar behavior if we make the S_H and S_h terms much smaller than P.

As a result, we can get an idea of the relative difference between the remote principal stresses and how they compare to P by studying the trajectory pattern. When we see a radial pattern, we know that that the pressure term is dominating, or that the remote stresses are close in size and magnitude to the pressure.

Figure 3: Plot of trajectories with $S_H = 125$, $S_h = 100$.

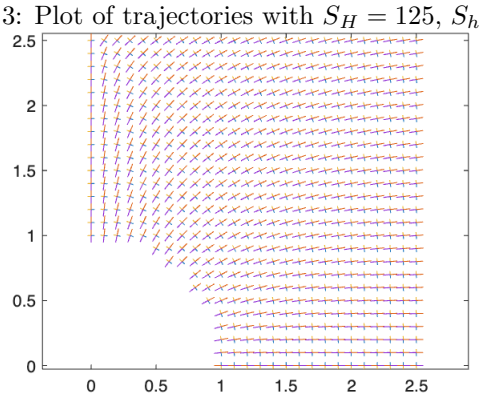


Figure 4: Plot of trajectories with $S_H = 1000$, $S_h = 100$.

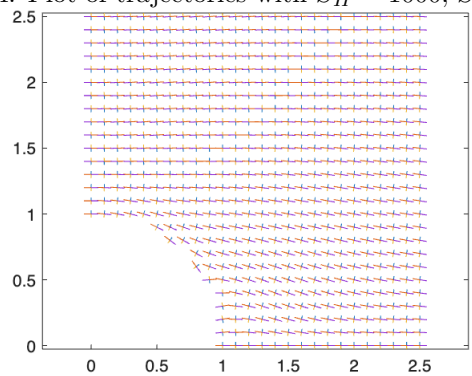
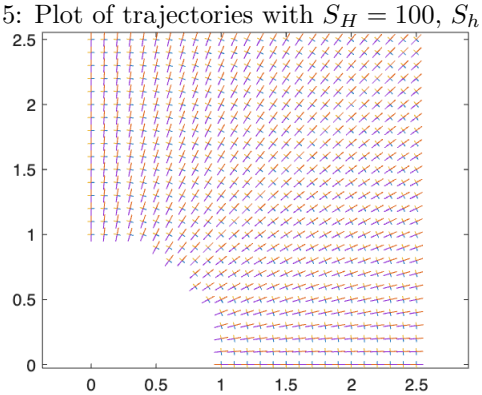


Figure 5: Plot of trajectories with $S_H = 100$, $S_h = 100$.



Problem 3

a - e: Write a MATLAB script (or function) that takes as input the six independent components of the stress tensor in a given Cartesian coordinate system

Please see stress.m for the implementation.

f. Now use your script to compute the normal component of the traction and the magnitude of the shear traction.

$$\sigma = \begin{pmatrix} -40 & 20 & -10 \\ 20 & -60 & 10 \\ -10 & 10 & -80 \end{pmatrix} \tag{30}$$

$$\bar{n} = (0.3015, 0.3015, 0.9045) \tag{31}$$

My script gives the normal component of the traction to be:

$$t(n)_n = -21.3775\hat{x} - 21.3775\hat{y} - 64.1324\hat{z} \tag{32}$$

and the magnitude of the shear component:

$$||t(n)_s|| = ||t(n) - t_n(n)|| \tag{33}$$

$$= 21.0854 \text{ MPa.} \tag{34}$$

g. the values and directions (as vectors in x, y, z) of the principal stresses (remember that in our convention is the greatest principal stress (the least negative)

In stress.m, we use the eig function to solve this problem. For the values in (30-31), we get back the values and direction of principal stress:

$$\sigma_1 = -27.4210, \text{with direction } (0.8593, 0.5070, -0.0670) \tag{35}$$

$$\sigma_2 = -62.0081, \text{with direction } (-0.3712, 0.7085, 0.6001) \tag{36}$$

$$\sigma_3 = -90.5709, \text{with direction } (0.3517, -0.4908, 0.7971) \tag{37}$$